

low walls rather than bouncing of them, and negotiate around items cluttering the floor. These can even escape if caught in a particularly difficult nook under a table.

The next generation of housekeeping robots promise to be a lot smarter. A robot designed to demonstrate cognitive systems that self-understand and self-extend (CogX) is one such example. This robot is equipped with probabilistic reasoning and planning capability to exploit facts or pieces of knowledge that it detects. For instance, if you ask this robot to find a pack of cornflakes, it will exploit the knowledge that there is a greater probability that someone has left it in the kitchen and start its search from there.

Technology like this can make robots far more intelligent than they are now, and more efficient in cases like the one mentioned above.

Step two: understanding emotions

We finally have the technology to ensure that even if your spouse doesn't understand your feelings, you can count on your robot. Computer Expression Recognition Toolbox (CERT) is a complete system for fully automated facial expression recognition. The CERT system helps detect spontaneous facial expressions, including automated discrimination of posed vs genuine expressions of pain, automated detection of driver drowsiness, adaptive tutoring systems and intervention for children with autism.

CERT also has its successor technology in the market, called FACET Vision by Emotient. Emotient API provides the ability to analyse emotional response of users, detecting everything from joy, surprise and anger to complex ones like frustration and confusion. It can even identify blends of two or more emotions. It does this by using Emotient API Facial Action Units (AUs). These action units are essentially elementary facial muscle movements that are detected and monitored by the robot or application to identify what kind of expression is being shown by the human being. Emotient AUs are factors used to detect over 28 human actions, from raising an eyebrow to a jaw drop. Emotient

Home automation gets a body

Social robots. The winner of this year's Oracle IoT Developer global challenge, BOT-SO, is an interesting robot for home security. Developed by Kolkata-based EdifiXio India, it can send photos, videos and other data as direct messages to its users via Twitter. The first prototype was made from easily-available components like Raspberry Pi Model B, camera module, micro servomotor, temperature sensor, motion detector and USB soundcard. "BOT-SO is in its very early stages, and would act as an open source reference platform for developing an intelligent robot," says Debraj of EdifiXio. The name BOT-SO is derived from the words 'social robot,' since it communicates through social media.

Furniture that changes shape. Roombots are mobile, adaptive, self-assembling, self-reconfigurable modular robots that could roll out and change their shape to a chair or table or box, or even disassemble, depending on user's instructions. Each Roombot comes with two connected spheres and four motors; like Lego bricks, it connects to other Roombots (on their own, unlike Lego bricks) to create the furniture you desire.



Roombots in action



Bob the Robot

Environment aware. Developed by the University of Birmingham, Bob is a tall robot that builds a 3D image of a room by using 3D sensors to scan. This robot then compares it with the previous image of the same room to detect any changes. If the settings of the room have changed (like a door being opened, or change in position of furniture), it immediately informs the concerned authorities. Bob also asks for help when stuck at a place. It charges itself when battery is low.

recently raised over six million US dollars for its facial recognition technology.

Affectiva, a company that grew out of MIT's Media Lab, develops Affdex, another emotion measurement technology that competes with FACET Vision. Affdex SDK enables any connected device, even a smartphone, to be able to read emotions and expressions. The data for such systems is gained through 3D cameras mounted on the robots, and through the use of sensor arrays on the robots' bodies. One example of a much-hyped idea for a successful application of this technology is to create robotic pets that help you in times of mental difficulty. In short, a robot can be your friend in need.

Step three: replace their animals

Robotics is not always about humanoid form-factors. Some robot concepts and designs that came up recently were inspired by fishes, birds and animals. Jayakrishnan of Asimov Robotics finds



JIBO helping in the kitchen

these models inspired by nature very intriguing. These robotic pets are also known as companion robots. They might be an emotional companion to some, or a utility companion to others.

PARO is one such high-tech interactive robot developed by AIST, a Japanese industrial automation company. It allows the benefits of animal therapy to be administered to people in environments where live animals present treatment or logistical difficulties. Think of it as your Tamagotchi (a hand-held digital pet created in Japan) with a body. Robotic pets use interactive and adaptive computing technology to use the input gained from their sensors to simulate